

SecureMailScope

SIH26159 — AI-Assisted Cryptographic Security Posture Assessment

Scan: SMS-PCAP-20260930-153342-acac | Mode: REAL_PCAP_SCAPY
Generated: 2026-09-30T10:03:43.209239+00:00

Overall posture: 68/100 — HIGH RISK

Sessions: 6 | Secure/Encrypted sessions: 5 | Findings: 9 | Critical: 1 | High: 4 | Medium: 4 | Low: 0

Session Inventory

Session	Protocol	TLS	Cipher	Certificate	Packets
imap-2	IMAP	TLS 1.0	TLS_RSA_WITH_RC4_128_SHA	not-observed	29-42
imap-5	IMAP	TLS 1.2	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	not-observed	71-84
pop3-3	POP3	PLAINTEXT	NONE	not-observed	43-56
smtp-0	SMTP	TLS 1.3	TLS_AES_256_GCM_SHA384	not-observed	1-14
smtp-1	SMTP	TLS 1.1	TLS_RSA_WITH_3DES_EDE_CBC_SHA	not-observed	15-28
smtp-4	SMTP	TLS 1.2	TLS_RSA_WITH_AES_256_CBC_SHA	not-observed	57-70

Findings

ID	Severity	Session	Finding	Evidence
TLS-001	HIGH	imap-2	Deprecated TLS version negotiated	Packets 29-42: negotiated TLS version = TLS 1.0
CRYPTO-002	HIGH	imap-2	Weak / deprecated cipher suite	Packets 29-42: cipher suite = TLS_RSA_WITH_RC4_128_SHA
CRYPTO-003	MEDIUM	imap-2	Forward secrecy not observed	Packets 29-42: key exchange = RSA
TLS-004	CRITICAL	pop3-3	Email session observed without TLS protection	Packets 43-56: email payload observed before any TLS handshake
CRYPTO-005	MEDIUM	smtp-0	Forward secrecy not observed	Packets 1-14: key exchange = Unknown
TLS-006	HIGH	smtp-1	Deprecated TLS version negotiated	Packets 15-28: negotiated TLS version = TLS 1.1
CRYPTO-007	HIGH	smtp-1	Weak / deprecated cipher suite	Packets 15-28: cipher suite = TLS_RSA_WITH_3DES_EDE_CBC_SHA
CRYPTO-008	MEDIUM	smtp-1	Forward secrecy not observed	Packets 15-28: key exchange = RSA
CRYPTO-009	MEDIUM	smtp-4	Forward secrecy not observed	Packets 57-70: key exchange = RSA

AI-Assisted Insights

- The evidence contains 1 critical finding(s); these should be addressed first.
- Key Exchange controls are the dominant source of observed cryptographic risk in this capture.
- Isolation Forest highlights imap-5 as the highest-anomaly session with score 1.0.
- Plaintext email access/submission is a direct transport-security concern and creates the strongest evidence-backed exposure in the demonstration capture.
- Risk conclusions are derived from observed protocol/cryptographic evidence; anomaly scores identify unusual profiles and do not by themselves prove compromise.

Recommendations

- Require TLS-protected email access/submission and disable plaintext access where policy permits.
- Disable TLS 1.0/1.1 and enforce TLS 1.2+; prefer TLS 1.3.
- Remove RC4/3DES and permit modern AEAD suites such as AES-GCM or ChaCha20-Poly1305.
- Prefer ephemeral ECDHE/X25519 key exchange to provide forward secrecy.

Integrity Evidence

Final evidence hash: 97a64ef10ff2e6472e0f10889006e2f3c5fcb0502d2056916e9883a459bc6e7c

Final audit record hash: e854c2c8011d6af6df137aa572077b4f40046a5d542fa18bac20ddaab75a8722

The current prototype uses a chained SHA-256 audit ledger. A production deployment can anchor the final evidence hash to a permissioned ledger.